PSet 1: The Price of Housing

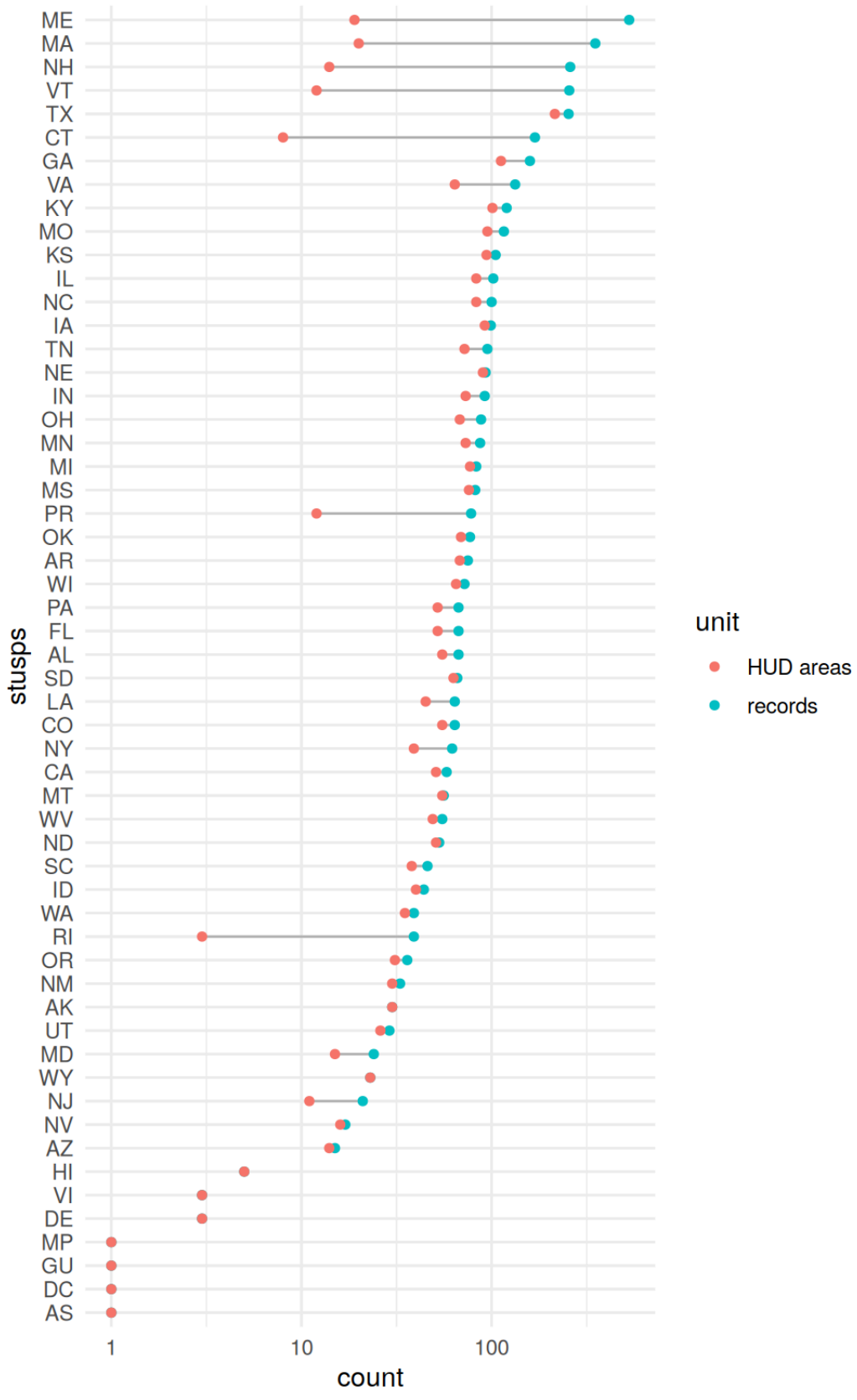


Figure 1

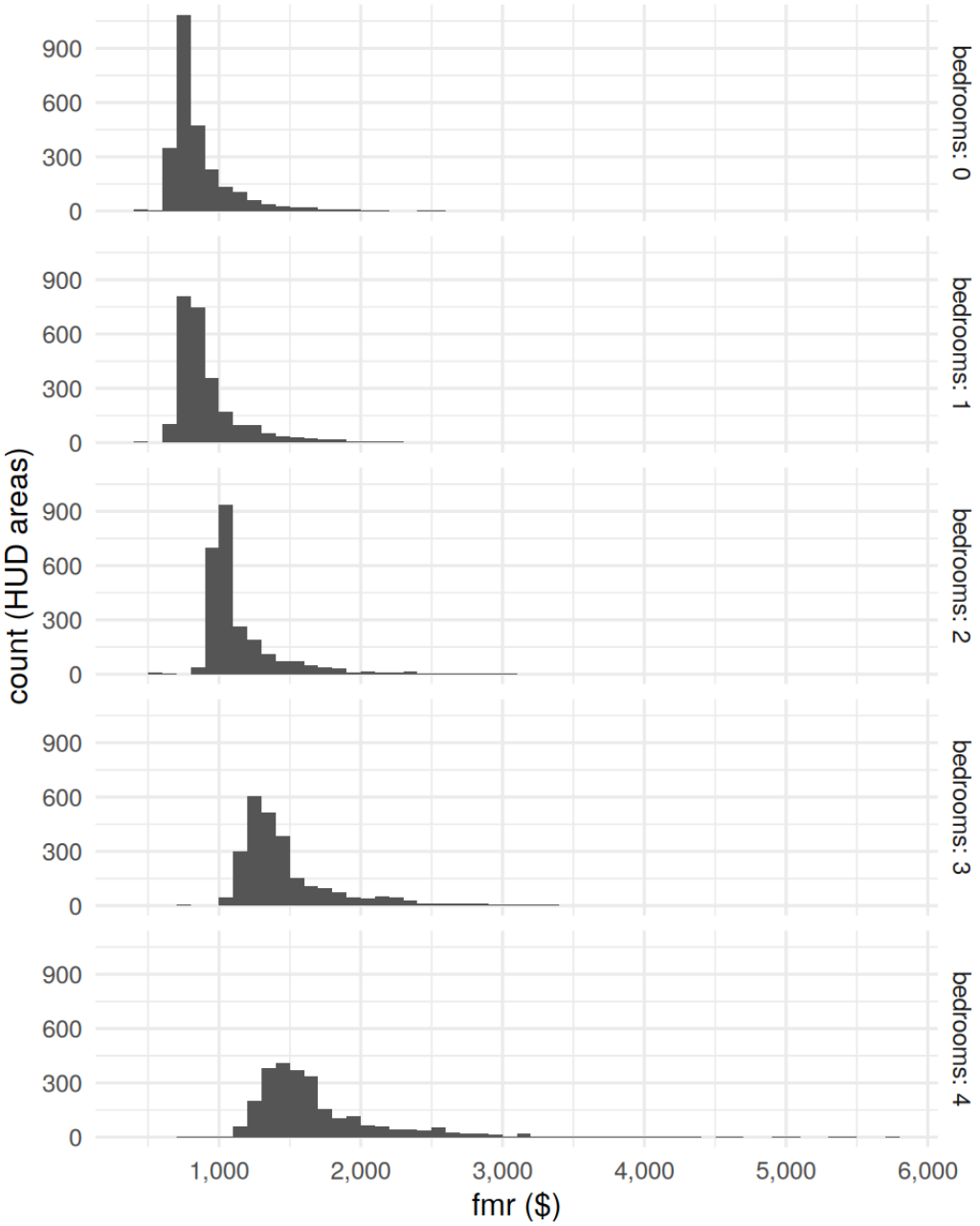

count (HUD areas)
900
600
300
0
bedrooms: 0
bedrooms: 1
bedrooms: 2
bedrooms: 3
bedrooms: 4
1,000
2,000
3,000
4,000
5,000
6,000
fmr ($)

Figure 2

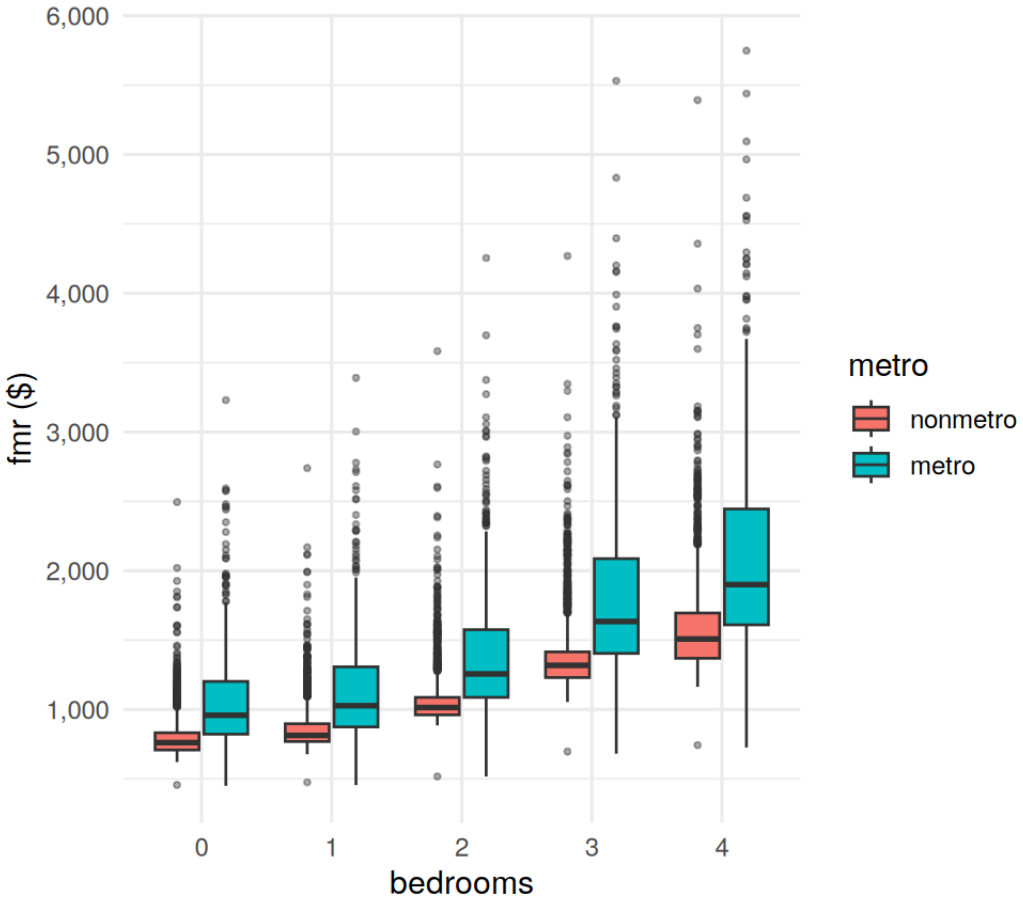

6,000
5,000
4,000
3,000
2,000
1,000
fmr ($)
0
1
2
3
4
bedrooms
metro
nonmetro
metro

Figure 3

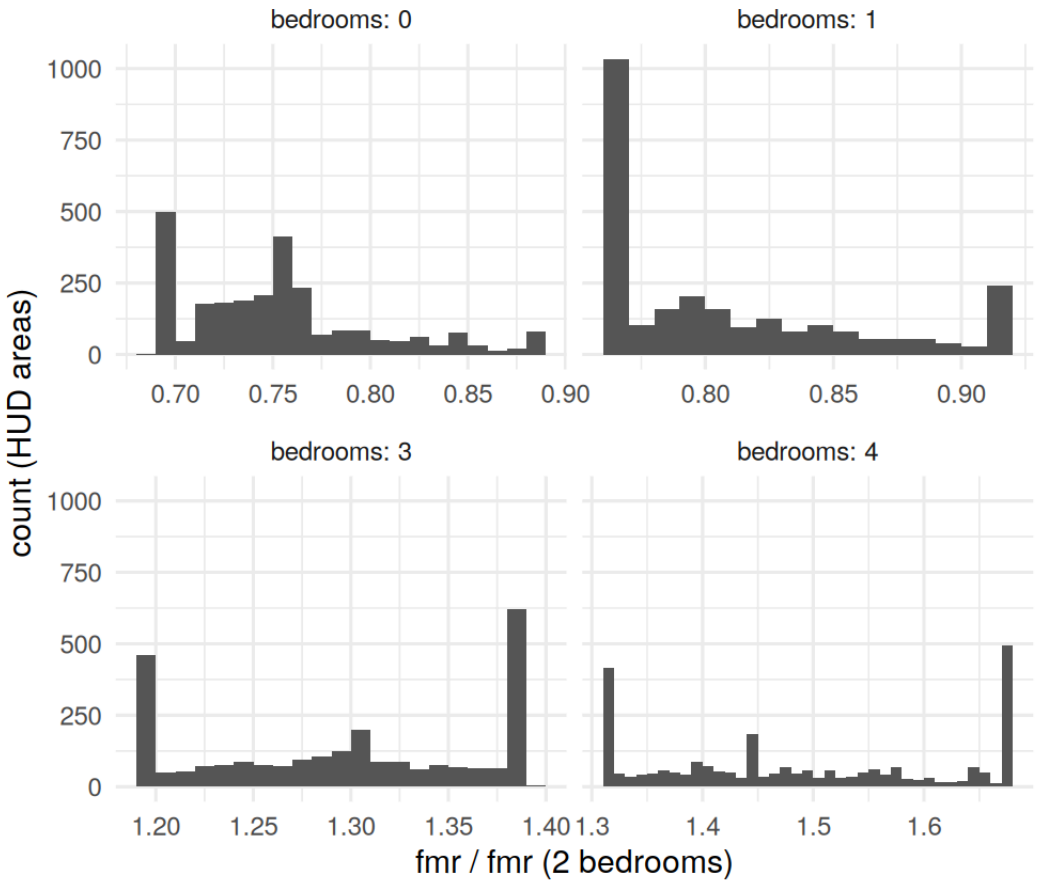

bedrooms: 0
bedrooms: 1
bedrooms: 3
bedrooms: 4
count (HUD areas)
1000
750
500
250
0
0.70
0.75
0.80
0.85
0.90
1.20
1.25
1.30
1.35
1.40
1.3
1.4
1.5
1.6
fmr / fmr (2 bedrooms)

Figure 4

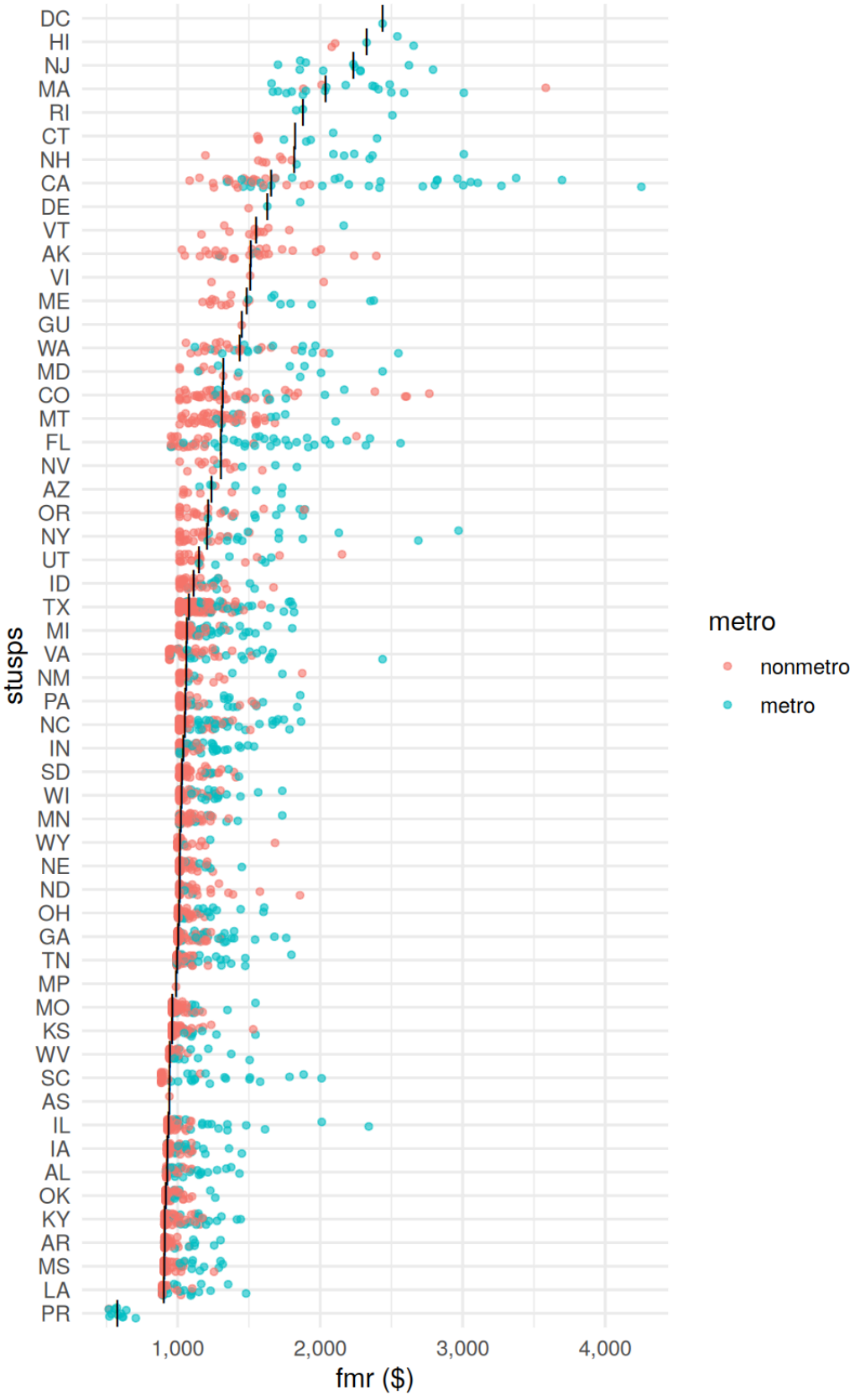

DC
HI
NJ
MA
RI
CT
NH
CA
DE
VT
AK
VI
ME
GU
WA
MD
CO
MT
FL
NV
AZ
OR
NY
UT
ID
TX
MI
VA
NM
PA
NC
IN
SD
WI
MN
WY
NE
ND
OH
GA
TN
MP
MO
KS
WV
SC
AS
IL
IA
AL
OK
KY
AR
MS
LA
PR
stusps
1,000
2,000
3,000
4,000
fmr ($)
metro
nonmetro
metro

Figure 5


4,000
3,000
2,000
1,000
fmr ($)
1,000
100,000
10,000,000
pop2024
metro
nonmetro
metro

Figure 6

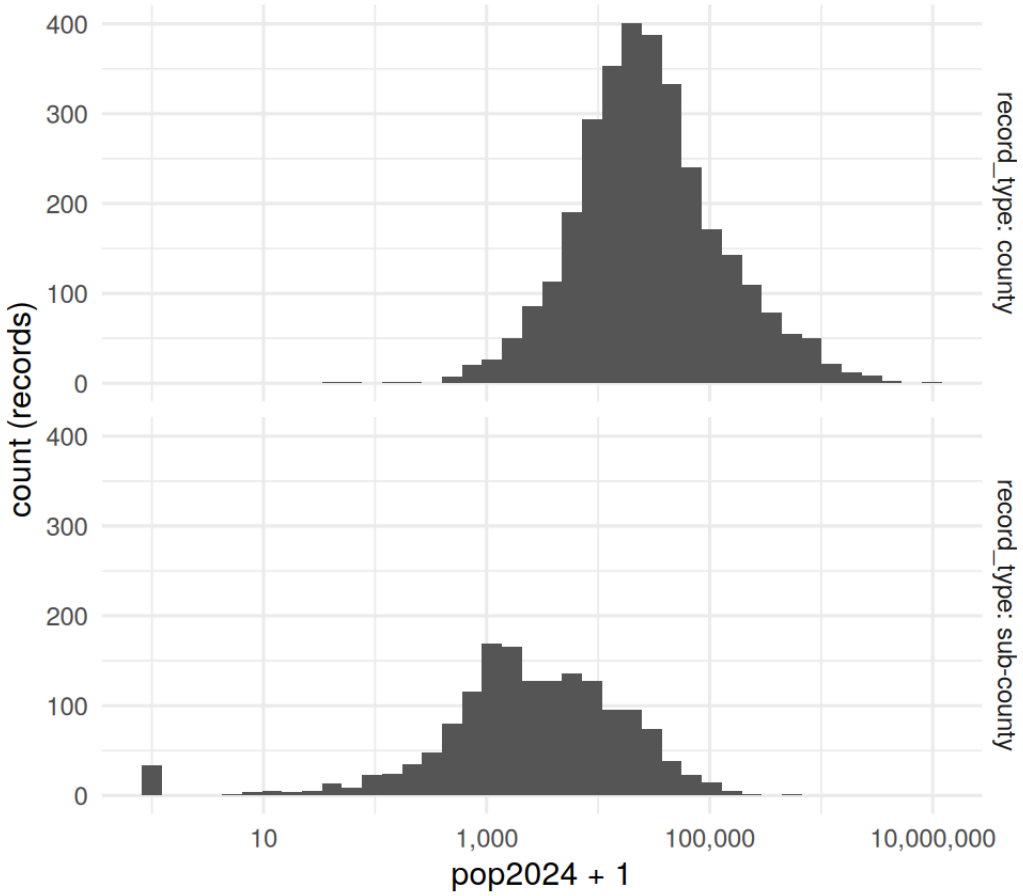


Figure 7

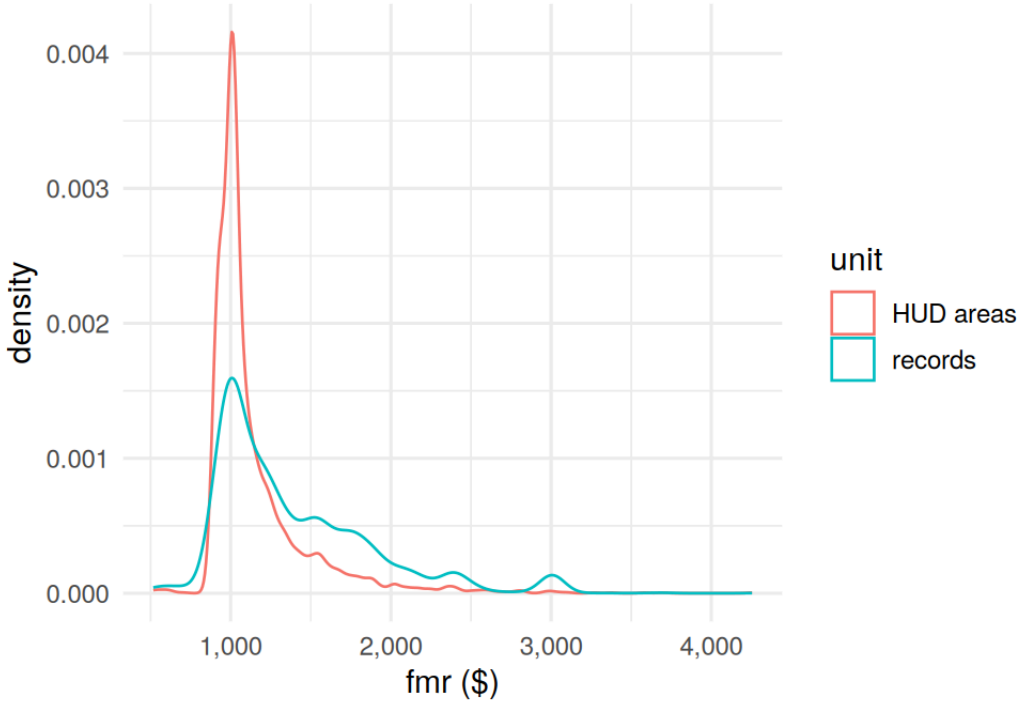


Figure 8

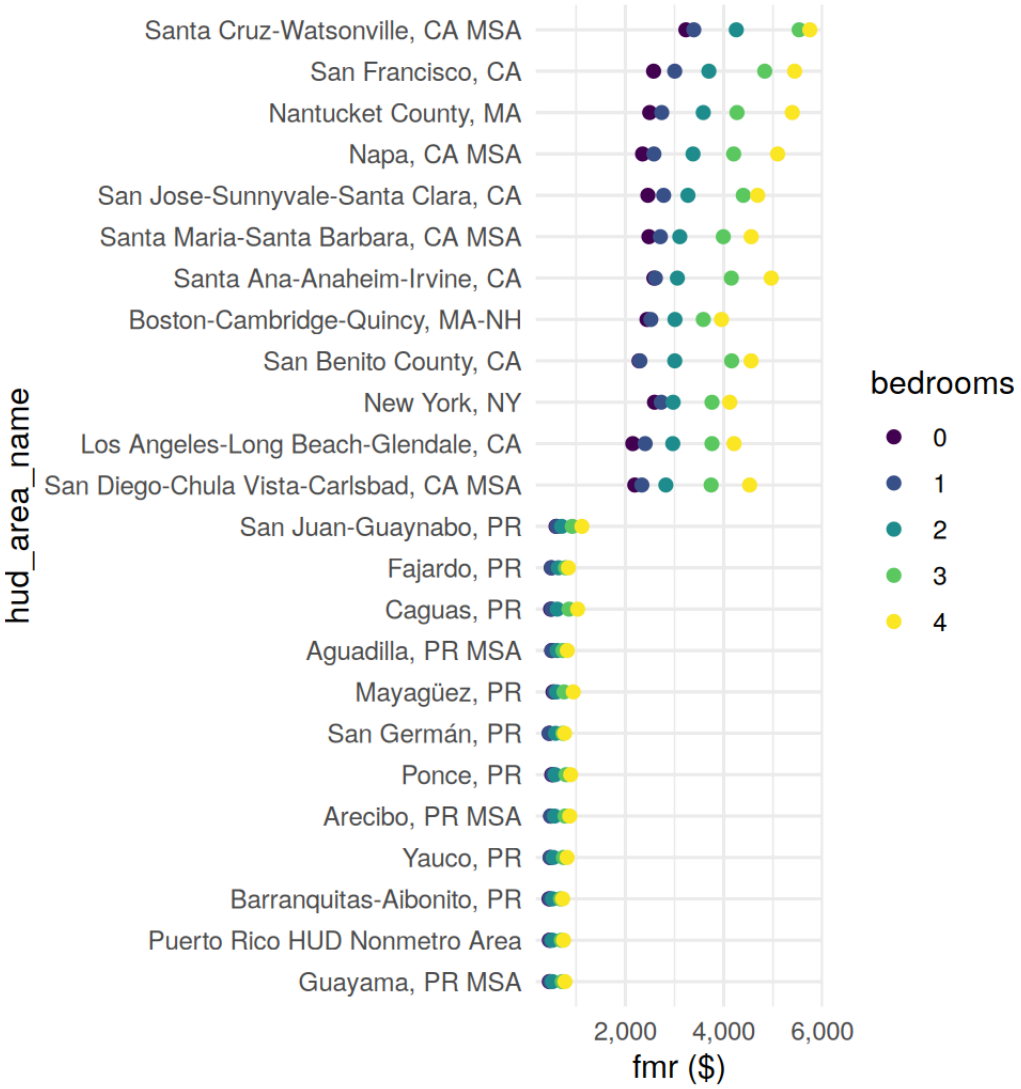

Santa Cruz-Watsonville, CA MSA
San Francisco, CA
Nantucket County, MA
Napa, CA MSA
San Jose-Sunnyvale-Santa Clara, CA
Santa Maria-Santa Barbara, CA MSA
Santa Ana-Anaheim-Irvine, CA
Boston-Cambridge-Quincy, MA-NH
San Benito County, CA
New York, NY
Los Angeles-Long Beach-Glendale, CA
San Diego-Chula Vista-Carlsbad, CA MSA
San Juan-Guaynabo, PR
Fajardo, PR
Caguas, PR
Aguadilla, PR MSA
Mayagüez, PR
San Germán, PR
Ponce, PR
Arecibo, PR MSA
Yauco, PR
Barranquitas-Aibonito, PR
Puerto Rico HUD Nonmetro Area
Guayama, PR MSA
hud_area_name
2,000
4,000
6,000
fmr ($)
bedrooms
0
1
2
3
4

Figure 9

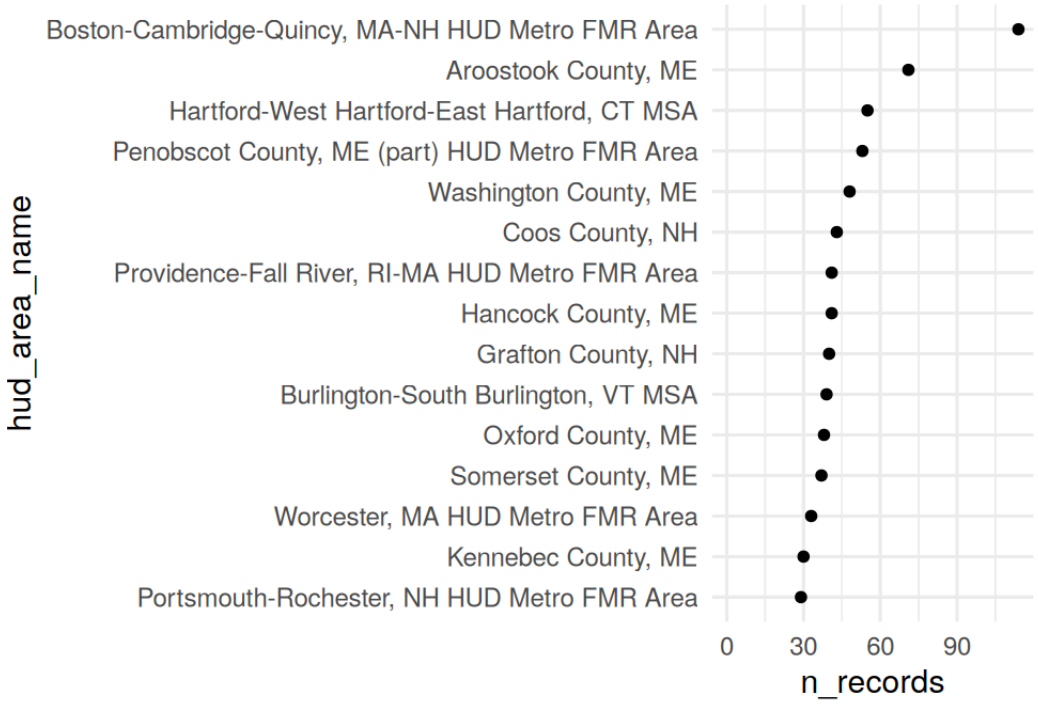

Boston-Cambridge-Quincy, MA-NH HUD Metro FMR Area
Aroostook County, ME
Hartford-West Hartford-East Hartford, CT MSA
Penobscot County, ME (part) HUD Metro FMR Area
Washington County, ME
Coos County, NH
Providence-Fall River, RI-MA HUD Metro FMR Area
Hancock County, ME
Grafton County, NH
Burlington-South Burlington, VT MSA
Oxford County, ME
Somerset County, ME
Worcester, MA HUD Metro FMR Area
Kennebec County, ME
Portsmouth-Rochester, NH HUD Metro FMR Area
hud_area_name
0
30
60
90
n_records

Figure 10

Data checks behind the figures

Checks on the data behind the figures. A record is one `fips`, an area is one `hud_area_code`.

```
rows: 23820  columns: 11
records (fips): 4764
HUD areas: 2606
states / territories: 56
bedroom sizes: 0, 1, 2, 3, 4
```

```
# A tibble: 1 x 2
  column           n_missing
  <chr>                <int>
1 county_town_name     15795
```

```
# A tibble: 3 x 3
  record_type missing_town_name     n
  <chr>       <lgl>             <int>
1 county      FALSE                 2
2 county      TRUE               3159
3 sub-county  FALSE              1603
```

```
fully duplicated rows: 0
duplicated fips + bedrooms: 0
records without every bedroom size: 0
records where fmr ever goes down as bedrooms go up: 0
```

```
# A tibble: 1 x 2
  n_fmr     n
  <int> <int>
1     1 13030
```

```
# A tibble: 1 x 4
  areas one_record two_or_more most_records
  <int>      <int>       <int>        <int>
1  2606       2325         281          114
```

```
# A tibble: 1 x 1
  areas_in_more_than_one_state
                         <int>
1                           36
```

```
# A tibble: 2 x 3
  metro    code_start     n
  <fct>    <chr>      <int>
1 nonmetro NCNTY       2788
2 metro    METRO       1976
```

```
# A tibble: 4 x 3
  stusps metro        n
  <chr>  <fct>    <int>
1 ME     nonmetro     8
2 ME     metro        3
3 NH     nonmetro    17
4 VT     nonmetro     6
```

```
# A tibble: 6 x 3
  stusps countyname       county_town_name
  <chr>  <chr>            <chr>
1 ME     Aroostook County Penobscot Indian Island Reservation
2 ME     Franklin County  West Central Franklin UT
3 ME     Hancock County   Marshall Island UT
4 ME     Hancock County   Northwest Hancock UT
5 ME     Knox County      Criehaven UT
6 ME     Knox County      Muscle Ridge Island UT
```

```
# A tibble: 6 x 3
  stusps record_type     n
  <chr>  <chr>       <int>
1 CT     sub-county    169
2 MA     sub-county    351
3 ME     sub-county    529
4 NH     sub-county    259
5 RI     sub-county     39
6 VT     sub-county    256
```

```
# A tibble: 5 x 5
  bedrooms   min median  mean   max
     <dbl> <dbl>  <dbl> <dbl> <dbl>
1        0   450   933   1050  3230
2        1   455   978.  1125  3390
3        2   517  1228   1388  4254
4        3   681  1611   1784  5531
5        4   726  1850.  2028  5748
```

```
# A tibble: 5 x 2
    fmr     n
  <dbl> <int>
1  1014   372
2   909    98
3  1002    67
4   961    66
5   962    62
```

```
# A tibble: 7 x 4
  stusps hud_area_name                                              fmr    pop
  <chr>  <chr>                                                    <dbl>  <dbl>
1 DC     Washington-Arlington-Alexandria, DC-VA-MD HUD Metro FMR A~  2438 6.16e6
2 AS     American Samoa                                               942 5.55e4
3 GU     Guam                                                        1449 1.59e5
4 MP     Northern Mariana Islands                                     988 5.39e4
5 VI     St. Croix Island, VI                                        1237 5.32e4
6 VI     St. John Island, VI                                         2025 4.20e3
7 VI     St. Thomas Island, VI                                       1509 5.12e4
```